

Mist Maker(Ultrasonic Mist Generation System using Piezoelectric Transducer)

Aim

Basic understanding about LT-Spice

LT-Spice simulation installation instructions

Connections - Circuit Diagram - Detailed Steps

Task - Exercise

Aim:

To design and simulate a piezoelectric-based mist maker system that generates fine mist using high-frequency vibrations.

LT-Spice – Introduction

LT-Spice is an open source, high-performance simulation tool developed

by Analog Devices. It is used to simulate, design, and test electronic circuits

before physically building them.

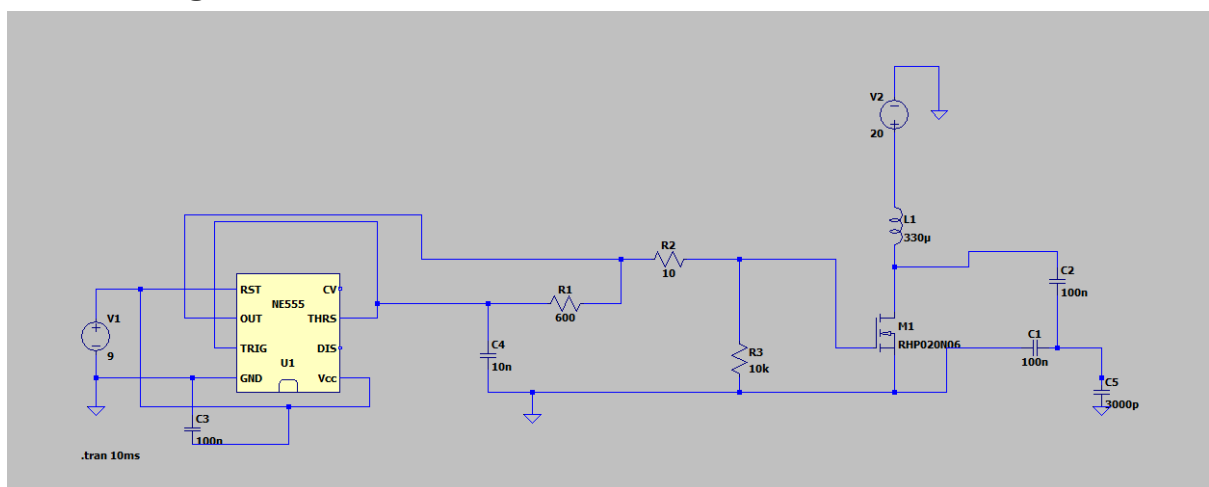
Why LT-Spice is important? (Theory)

LT-Spice provides a safe learning environment where peoples can freely experiment with electronic circuits without risk of damaging physical components. It helps them to understand circuit behaviour by allowing real time visualization of voltage, current, and other parameters. By simulating before building, Peoples save time and cost as they as they do not need to buy multiple components or PCBs initially. LT-Spice also encourages peoples to develop strong problem -solving skills, teaching them how to identify and fix circuit issues logically. It offers valuable hands-on simulation experience that connects theoretical concepts learned in classrooms with practical real-world electronics. Most importantly, learning LT-Spice gives peoples and industry-relevant skill, preparing them for careers in research, development, and engineering industries.

LT-Spice Simulation Installation Instructions

- 1) Open your web browser (Such as Chrome, Firefox, or Edge).
- 2) In the search bar, type LT-Spice and press Enter.
- 3) Click on the official link from Analog devices, usually titled something like “LTspice” | Analog devices”. Or use this hyperlink to download [LT-Spice](#).
- 4) On the LTspice page, look for a Download button or section.
- 5) Choose the correct version for your operating system (Windows, macOS, etc.).
- 6) Download the installer file to your computer.
- 7) Once the download is complete, open the installer file.
- 8) Follow the on-screen instructions to install LT-Spice:
 - Accept the License Agreement
 - Choose the installation folder (or use the default)
 - Complete the installation process.
- 9) After installation, open LT-Spice from your desktop or Start Menu.
- 10) Now are you ready to start using LT-Spice for circuit simulation.

Circuit Diagram:



Detailed Connection Steps:

Step 1: Open LT-Spice

- Open LT-Spice on your system.
- Go to **File → New Schematic**

Step 2: Place required Components

Press P (Pick Component) and place the following

Component	Symbol in LT-Spice	Search Keyword
Voltage Source	V	Voltage
NE555 Timer IC	NE555	NE555
Resistor	R	Res
Capacitor	C	Cap
Inductor	L	Ind
Mosfet(NMOS)	M	Nmos
Ground	GND	Gnd

Step 3: Place and Configure Voltage Sources

- Place **two voltage sources**:
 - **V1 = 9V** → Supply for NE555 circuit
 - **V2 = 20V** → Power stage (mist generation section)
- Right-click and rename:
 - First source → **V1**
 - Second source → **V2**

Step 4: NE555 Timer Connections (Oscillator Section)

1. Connect **V1 (9V)** to:
 - **Pin 8 (Vcc)** of NE555
 - **Pin 4 (Reset)**
2. Connect **Pin 1 (GND)** to Ground.
3. Short the following pins:
 - **Pin 2 (Trigger)** and **Pin 6 (Threshold)**
4. Connect:

- **C3 (100nF)** between **Vcc and GND** (decoupling capacitor)

5. Connect:

- **C4 (10nF)** from **Pin 5 (Control Voltage)** to Ground

6. Output:

- **Pin 3 (OUT)** goes to resistor network (R1)

Step 5: Signal Conditioning Network

1. Connect **R1 (600Ω)** from NE555 output to next stage
2. Connect **C4 (10nF)** from the R1 node to Ground
3. Connect **R2 (10Ω)** in series after R1
4. Connect **R3 (10kΩ)** from MOSFET gate node to Ground (pull-down resistor)

Step 6: MOSFET Switching Stage

1. Place **NMOS (M1: RHP020N06)**
2. Connections:
 - **Gate (G)** → From R2 output
 - **Source (S)** → Ground
 - **Drain (D)** → Inductor + output node

Step 7: Inductor and Power Stage

1. Connect **V2 (20V)** positive to **Inductor L1 (330μH)**
2. Connect other side of **L1** to:
 - **Drain of MOSFET (M1)**
 - Output network

Step 8: Output and Resonant Section (Mist Generation Part)

1. Connect **C1 (100nF)** from output node to ground
2. Connect **C2 (100nF)** across output node (filtering)

3. Connect **C5 (3000pF)** from output node to ground

Step 9: Ground Connections

Ensure all grounds are common:

- NE555 GND
- MOSFET Source
- Capacitor grounds
- Voltage source negatives

Step 10: Wiring

- Press **F3** to wire connections
- Carefully connect all nodes exactly as per diagram
- Avoid floating nodes

Step 11: Set Component Values

Component	value
V1	9V
V2	20V
R1	600Ω
R2	10Ω
R3	10kΩ
C3	100nF
C4	10nF
C1, C2	100nf
C5	3000pf
L1	330μH
MOSFET	RHP020N06

Step 12: Simulation Setup

- Press **S → Transient Analysis**
- Set:
 - **Stop Time = 10ms**

- Place command:

.tran 10ms

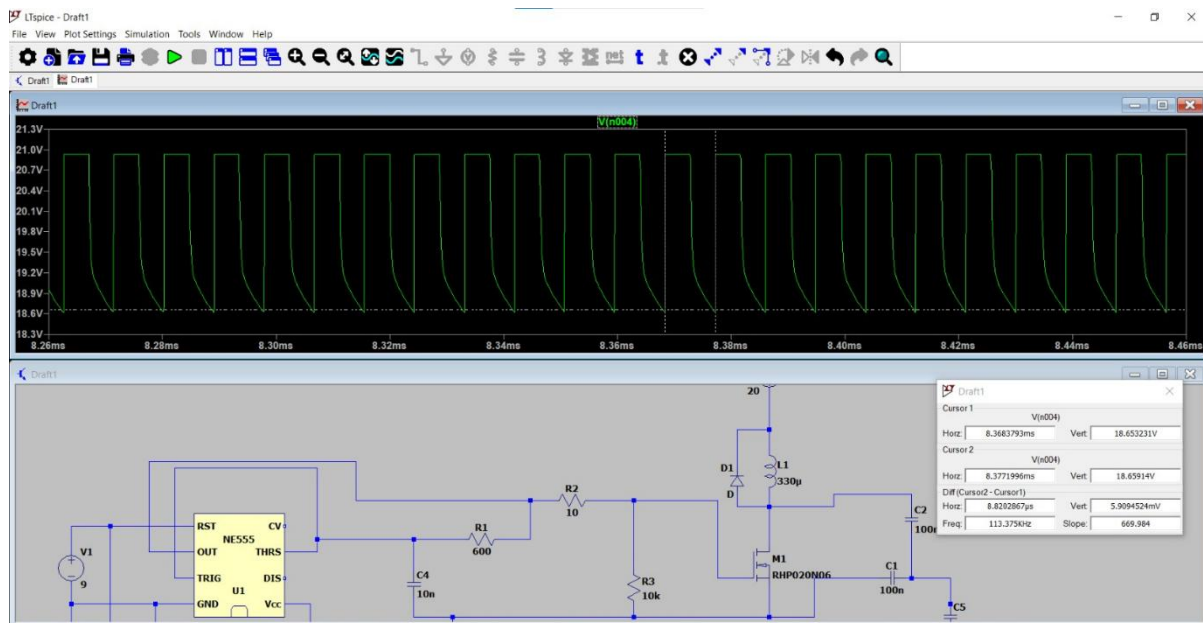
Step 13: Run Simulation

- Click **Run**
- Observe:
 - Output waveform at MOSFET drain
 - Oscillation frequency from NE555
 - Resonant behavior at output

Step 14: Verification

- Ensure oscillation is stable
- Check high-frequency switching at MOSFET
- Confirm waveform suitable for piezoelectric excitation

Expected Output:



Tasks:

1. Oscillator Frequency Setup

Design the NE555 timer circuit to generate high-frequency pulses suitable for piezoelectric operation. Verify the output waveform and measure the frequency to ensure it falls within the ultrasonic range required for mist generation.

2. MOSFET Switching Analysis

Simulate the MOSFET switching behavior driven by the NE555 output. Observe the gate signal and verify proper switching at high frequency without distortion or delay.

3. Output Waveform Observation

Measure and analyze the voltage waveform at the MOSFET drain. Confirm that high-frequency oscillations are generated and are suitable to drive the piezoelectric element.

4. Resonant Network Tuning

Modify capacitor values (C1, C2, C5) and observe how the output waveform changes. Identify the condition where resonance is achieved for efficient mist generation.

5. Supply Voltage Variation

Change the input voltage (e.g., 9V to 12V or 15V) and observe its effect on output amplitude and system performance. Analyze how voltage levels influence mist generation efficiency.